The Path Union Dimension Principle: Synthesis of Higher-Dimensional Manifolds via Uncountable Families of Lower-Dimensional Trajectories

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Abstract

We formalize the Path Union Dimension Principle (PUDP), establishing the topological conditions under which an uncountable family of lower-dimensional continuous trajectories completely covers a compact, path-connected region $R \subset \mathbb{R}^n$. We prove that if every point $x \in R$ intersects at least one admissible continuous trajectory Γ, the union of these subsets preserves the topological dimension and Lebesgue measure μ_n of the ambient space.

1 The Two-Dimensional Framework

Let $R \subset \mathbb{R}^2$ be a path-connected region. For any point $x \in R$, we define continuous sub-paths mapping from boundary reference points P and Q:

$$\alpha : [0, 1/2] \to R \quad \text{where} \quad \alpha(0) = P,\ \alpha(1/2) = x$$

$$\beta : [1/2, 1] \to R \quad \text{where} \quad \beta(1/2) = x,\ \beta(1) = Q$$

We define the concatenated path $\gamma(t) : [0, 1] \to R$ as:

$$\gamma(t) = \begin{cases} \alpha(t), & 0 \le t \le 1/2 \\ \beta(t), & 1/2 < t \le 1 \end{cases}$$

By double inclusion, the total area A operator maps smoothly onto the uncountable union:

$$R = \bigcup_{\gamma} \gamma([0, 1]) \implies A(R) = A\left(\bigcup_{\gamma} \gamma([0, 1])\right) \tag{1}$$

2 Generalization to n-Dimensions

Let R be a bounded region such that $\dim(R) = n$. Let Γ represent a continuous trajectory of lower dimension $\dim(\Gamma) = n - 1$. If every coordinate point $x \in R$ satisfies $x \in \Gamma$, then:

$$\bigcup_{\Gamma} \Gamma = R$$

Applying the n-dimensional Lebesgue measure μ_n to the manifold yields:

$$\boxed{\mu_n\left(\bigcup_{\Gamma} \Gamma\right) = \mu_n(R)} \tag{2}$$